

Niccolò Covoni

PHD STUDENT · UNIVERSITY OF ITALIAN SWITZERLAND · UNIVERSITY OF URBINO

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Area of Research _____

Philosophy of Information, Philosophy of Physics, Philosophy of Science,
and Metaphysics

Education _____

University of Svizzera Italiana USI

Lugano, Switzerland

PHD IN PHILOSOPHY

01/11/2023 - present

- Supervisor: Claudio Calosi

University of Urbino "Carlo Bo - Department of Pure and Applied Sciences

Urbino, Italy

PHD IN RESEARCH METHODS IN SCIENCE AND TECHNOLOGY

01/11/2023 - present

- Title of project: The Reality of Entropy: foundational study of entropy and information
- Supervisor: Vincenzo Fano

University of Firenze

Florence, Italy

MS IN LOGIC, PHILOSOPHY AND HISTORY OF SCIENCE

2020 - 2023

- Title of dissertation: Methods of Abstraction in Information Theory The Case of Quantum Like Machine Learning
- Supervisor: Silvano Zipoli Caiani
- Co-Supervisor: Giuseppe Sergioli
- Graduation Mark: 110/110 cum Laude

University of Pisa

Pisa, Italy

BS IN PHILOSOPHY

2017 - 2020

- Title of dissertation: Considerations on the notion of Birkhoff-Von Neumann's Quantum Logic
- Supervisor: Luca Bellotti
- Graduation Mark: 107/110

Publications _____

PUBLISHED

- Niccolò Covoni, Marco Sanchioni. 2026. Relational Information Towards a New Kind of Information in Quantum Mechanics in *Foundations of Physics*, 56(1): 14.
- Niccolò Covoni, Vincenzo Fano, Marco Sanchioni. 2025. Relational Quantum Mechanics and (Soft) Perspectivism in *Foundations of Physics*, 56(1): 6.
- Niccolò Covoni, Carlo Rovelli. 2025. Tractatus Quanticus, <https://arxiv.org/abs/2512.06034>
- Niccolò Covoni, Gino Tarozzi. 2025. Tra realismo e neopositivismo: l'eredità di Ludovico Geymonat e Evandro Agazzi nella filosofia della scienza italiana in *Bíos theôrêtikós e bíos praktikós. Studi offerti a Fabio Minazzi per i suoi settant'anni*, Mimesis Edizioni, Milano.
- Niccolò Covoni, Flavia Marcacci, Gino Tarozzi. 2025. From neo-idealistic historicism to Geymonat's materialistic and Agazzi's objectivistic realism: the roots of the Italian philosophy of science, in *BIOETHICS UPDATE*, 11(2), <https://doi.org/10.24875/BUP.25000013>.
- Niccolò Covoni. 2025. From Shannon to Von Neumann: A partial understanding of Information, in *Atti del XLIV Congresso Nazionale SISFA - Firenze 2024*, 43(10), 89-94.
- Niccolò Covoni, Giovanni Galli. 2025. A (quasi)-Mathematical Theory of Public Language, in *Proceedings 46th International Wittgenstein Symposium 2025*.
- Niccolò Covoni, Giovanni Macchia, Davide Pietrini and Gino Tarozzi. 2025. Non-standard realistic models of quantum phenomena and new forms of complementarity, in *Models and Representations in Science, edited by Hans-Peter Grosshans. Comptes Rendus de l'Académie Internationale de Philosophie des Sciences* 3, 39-54.
- Niccolò Covoni, Alberto Corti, Vincenzo Fano. 2024. A Model-Based form of Naturalised Metaphysics, *Revista Portuguesa de Filosofia*, 80(4), 783-818.
- Niccolò Covoni, Gino Tarozzi. 2024. Empiricism and Realism Reconciled in Agazzi's Conception of Scientific Objectivity, *Distinctio: Journal of Intersubjective Studies*, Zagreb, 3(1), 47-55.
- Niccolò Covoni, Silvano Zipoli Caiani. 2024. Levels of Abstraction as family resemblances from the classical to the quantum mechanical representation of reality in *Proceedings 45th International Wittgenstein Symposium 2024 "Facets of Reality — Contemporary Debates"*
- Marco Ciardi, Niccolò Covoni. 2024. La profezia di Ada Lovelace. L'intelligenza artificiale prima di Turing. On Filosofia dell'intelligenza artificiale. Sfide etiche e teoriche, Il Mulino, Bologna, ISBN: 978-88-15-38737-0
- Marco Ciardi, Niccolò Covoni. 2023. Dalla Mano Per Calcolare ai Calcolatori per la Mano. On L'Origine del Pensiero Numerico, Guide Museo e Istituto fiorentino di Preistoria, ISBN: 979-12-80445-06-3.

IN REVIEW

Niccolò Covoni, Hanna Podsiadkowska, Rafał Wieczorek. 2026. Segal Entropy and Relational Information

Alberto Corti, Niccolò Covoni, Vincenzo Nespeca, Marco Sanchioni. 2026. Progressive Scale Realism A naturalized realistic interpretation of Spin

Niccolò Covoni, Marco Sanchioni. 2026. Relationality without Agents: Quantum Correlations and the Limits of QBism

Niccolò Covoni, Marco Sanchioni. 2026. Progress Crucial Realism

Niccolò Covoni, Marco Sanchioni. 2026. Entanglement, Relative Facts, and the Meaning of Information in Relational Quantum Mechanics

Niccolò Covoni, Marco Sanchioni. 2026. Progress-Crucial Realism Toward a Selective Meta-Ontology

Niccolò Covoni, Marco Sanchioni. 2026. From “It from Bit” to Relational Quantum Mechanics: Wheeler, Rovelli, and the Relational Turn in Quantum Foundations

UNDER CONTRACT

Carlo Rovelli, Niccolò Covoni, Andrea Di Biagio. 2026. Tractatus Quanticus, John Wiley & Sons Ltd

Prizes & Grants

01/2025 - 03/2025	Scholarship from The QISS consortium , University of Western Ontario	€3,000
from 1/11/2023	PhD Scholarship founded by Department of Pure and Applied Sciences (DiSPeA) , University of Urbino ‘Carlo Bo’	€16,243,00 per year
2020-22	Scholarship at ”Percorso di Eccellenza Trasversale 2020/21” as Excellent Student , University of Florence	€4,000

Presentations

INVITED TALKS

- 23/04/2026. **Relational Information Towards a New Kind of Information in Quantum Mechanics.** Weekly Meeting Departamento de Fisica Teorica, Universidad Complutense de Madrid, Madrid, Spain.
- 19/04/2026. **Entanglement, fatti relativi e il significato di "informazione" nell' interpretazione relazionale della meccanica quantistica.** Il Paradosso della realta' fisica 1985 -225, University of Urbino, Cesena, Italy.
- 11/03/2026. **Relational Information Towards a New Kind of Information in Quantum Mechanics.** University of Jagellonian, Krakow, Poland.
- 05/03/2026. **Relational Information Towards a New Kind of Information in Quantum Mechanics.** Katedra Teorii Prawdopodobieństwa i Statystyki WMil UŁ wraz z Oddziałem Łódzkim PTM, University of Lodz, Lodz, Poland.
- 17/12/2025. **Once Again on von Neumann entropy: is Always The Case That von Neumann Entropy Correspond to Thermodynamic Entropy?.** Workshop: Philosophical Foundations of Thermodynamics, Ponteficia Universidad Católica De Valparaíso, Valparaíso, Chile.
- 20/11/2025. **Relational Information: Towards a New Type of Information in Quantum Mechanics.** Invited talk: Physics meets Philosophy in Vienna, Institute for Quantum Optics and Quantum Information, Vienna, Austria.

CONTRIBUTED PRESENTATIONS

- Relational Information Towards a New Kind of Information in Quantum Mechanics**
22/07/2026. Asian Philosophy of Science Association (APSA) Conference 2026, NTU Singapore, Singapore
- Segal Entropy and the notion of Information**
09/05/2026. Workshop: Perspectives on the Foundations of Information Theory and Statistical Mechanics Università della Svizzera Italiana, Lugano, Switzerland
- From 'It from Bit' to Relational Quantum Mechanics: Wheeler and Rovelli in Historical and Conceptual Perspective**
10/12/2025. Towards a Quantum Culture: The Different Interpretations of Quantum Mechanics, Workshop of the Italian Society for the History of Physics, Università Studi di Milano, Milano, Italy
- Relational Quantum Mechanics and (Soft) Perspectivism**
16/10/2025. Philosophy of Science Conference on the Occasion of the 140th Anniversary of Niels Bohr's Birth, Sofia, Bulgaria
- A (quasi)-Mathematical Theory of Public Language**
12/08/2025. 46th International Wittgenstein Symposium, Kirchberg am Wechsel, Austria

Progress Crucial Realism

11-13/06/2025. SILFS Postgraduate Conference, Turin, Italy

From Shannon to Von Neumann: A partial understanding of Information

18/09/2024. XLIV National Congress of SISFA, Firenze, Italy

The Forbidden Realm for the Maxwell's Demon: Information Entropy

09/09/2024. Synergia PhD Research Symposium - Charting New Frontiers in Research

Levels of Abstraction as family resemblances: from the classical to the quantum mechanical representation of reality

12/08/2024. 45th International Wittgenstein Symposium, Kirchberg am Wechsel, Austria

Exploring Quantum Information Processes through the Method of Abstraction: Quantum-Levels of Abstraction

04/08/2024. XXV World Congress of Philosophy, Rome, Italy

Fearing Bitcoin

03/08/2024. XXV World Congress of Philosophy, Society Section SILFS, Rome, Italy

Method of Abstraction: A Tentative Exploration of Quantum Information Processes

07/12/2023. Seminar organized by SYNERGIA the Research Group in Logic, Language, Cognition, History and Philosophy of Science, Università di Urbino Carlo BO, Urbino, Italy.

Contextuality of Abstraction

31/03/2023. Seminar organized by the professor of Philosophy of Mind Silvano Zipoli Caiani. University of Firenze, Firenze, Italy.

Una breve panoramica della Filosofia dell'informazione: dall'informazione classica all'informazione quantistica

10/11/2022. Seminar organized by the student university association Gavagai. Pisa, Italy.

POSTER PRESENTATIONS**Progress Crucial Realism**

29/08/2025. 10th biennial meeting of the European Philosophy of Science Association. Groningen, Netherlands.

Progress Crucial Realism

01/07/2025. Foundation2025. Gdańsk, Poland.

Shannon's approach to quantum information

14/10/2024. Probability in Computer Science (PICS) 2024. Copenhagen, Denmark.

PUBLIC PRESENTATIONS

Certo per Incerto: Quanta Informazione

20/04/2026. Facoltà di Teologia di Lugano. Lugano, Switzerland.

Teaching Experience

2026	Theory of Information , Philosophical Issues in Thermodynamics, Statistical Mechanics, and Information Theory	USI - Lugano
2025/2026	Philosophy of Language , Teaching Assistant in the course 'Philosophy of Language' taught by professor Marco Sanchioni	Sophia University Institute
09/05/2024	Introduction to Edutainment , Lecture for master students of Communication Science taught by professor Silvano Zipoli Caiani	University of Florence
02/2024 - 05/2024	Lectures about Epistemology , Teaching Assistant in the course 'Epistemology' taught by professor Gino Tarozzi	University of Urbino
15/12/2023	Introduction to Description Logic , Lecture for master students in Philosophy, in the course 'Web's Semantics' taught by professor Pierluigi Graziani	University of Urbino

Research Experience

07/26 - 08/26	Visting Student , Advisor: Professor. Eugene Chua, Nanyang Technological University Singapore	Singapore
02/26 - 04/26	Visting Student , Advisor: Professor. Hanna Podseǳkowska, University of Lodz -Faculty of Mathematics and Computer Science	Lodz, Poland
01/25 - 03/25	Visting Student , Advisor: Professor. Francesca Vidotto, University of Western Ontario - Department of Philosophy	London, ON, Canada
2022	Visting Student , Advisors: Professor. Roberto Giuntini, Professor Giuseppe Sergioli, University of Cagliari - Department of Pedagogy, Psychology, Philosophy	Cagliari, CA, Italy

Attendeed Events

CONFERENCES AND WORKSHOPS

Workshop: Exploring Quantum Technology and the Philosophy of Quantum Mechanics, 22/08/2025, TU Delft, Delft, Netherlands

QISS 2025 Conference, 14/04/2025-17/04/2025, Institute for Quantum Optics and Quantum Information (IQOQI), Wien, Austria

Workshop: Theoretical Equivalence and Related Topics, 08/04/2025, Università Svizzera Italiana, Lugano, Switzerland

4th Ca'Foscari-Lugano Workshop in Analytic Metaphysics, 16/12/2024, University Ca'Foscari, Venice, Italy

Workshop on Scientific Realism, 08/11/2024, University of Roma Tre, Rome, Italy

Ninth Annual Conference of the International Association for the Philosophy of Time, 24/06/2024-28/06/2024, University Svizzera Italiana, Lugano, Switzerland

Philosophy of Spontaneous Collapse, 20/05/2024-21/05/2024, University Svizzera Italiana, Lugano, Switzerland

Buchi, ombre, specchi, errori. Trent'anni di metafore nella filosofia di Achille Varzi, 02/05/2024, University Ca'Foscari, Venice, Italy

Variations in Philosophy of Mathematics, 13/04/2024-14/04/2024, University Svizzera Italiana, Lugano, Switzerland

Unification of Types and Multi-Level Modeling, 13/03/2024, King's College, London, United Kingdom

Formal Modeling of Ignorance, 07/09/2023 – 08/09/2023, Università degli Studi di Urbino Carlo BO, Urbino Italy

Triennial International Conference Logic, Philosophy of Science, and Global Challenges, 04/09/2023 – 07/09/2023, SILFS, Urbino, Italy

Linguaggio, Logica e Ontologia, 14/06/2023 – 15/06/2023, University of Pisa, Pisa, Italy

Information and Computing in Nature Philosophical Perspectives, 01/12/2022, Polish Academy of Arts and Sciences Commission for the Philosophy of Science, Online

Online Course 8 consciousness, 12/08/2022 – 19/08/2022, Steven Gouveia, Online

The 22nd Trends in Logic workshop, entitled Strong and Weak Kleene Logics, 18/07/2022 – 20/07/2022, ALOPHIS research group of the Department of Pedagogy, Psychological Sciences and Philosophy (Univestiy of Cagliari), Online

The manifest image and the scientific image: objects, properties and relations, 29/11/2021 – 30/11/2021, University of Firenze, Florence, Italy

The Nature of Quantum Objects, 11/11/2021 – 13/11/2021, University of Geneva, Geneva, Switzerland

PM65 An international workshop of logic to celebrate Pierluigi Minari's 65th birthday, 15/10/2021, University of Firenze, Florence, Italy

WINTER SCHOOLS

Probability in Computer Science (PICS) 2024 5th Ph.D. school in the series of Foundations of Programming and Software Systems, 14/10/2024 - 18/10/2024, It-Universitet, Copenhagen, Denmark, 2 ECTS.

SUMMER SCHOOLS

14th Ludwig Wittgenstein Summer School, 01/08/2023 – 05/08/2023, Austrian Ludwig Wittgenstein Society (ALWS), Kirchberg am Wechsel, Austria

XXV International Summer School in Philosophy of Physics, 06/06/2022 – 11/06/2022, Università degli Studi di Urbino Carlo BO, Urbino, Italy

12th Ludwig Wittgenstein Summer School, 03/08/2021 – 07/08/2021, Austrian Ludwig Wittgenstein Society (ALWS), Klagenfurt am Wörthersee, Austria

XXV International Summer School in Philosophy of Physics, 07/06/2021 – 11/06/2021, Università degli Studi di Urbino Carlo BO, Urbino, Italy

Service to the Profession _____

ACTIVE ORGANIZATION OF EVENTS

XXXI International Summer School in Philosophy of Physics - History and Philosophy of the Wave Function

15/06/2026 - 19/06/2026

Organiser, Università degli Studi di Urbino Carlo BO, Urbino, Italy

Edutainment and Social 2026

16/05/2024 - 17/05/2024

Conference organiser, <https://ludicomix.it/edutainment-social/> Empoli, Italy

Workshop: Perspectives on the Foundations of Information Theory and Statistical Mechanics

07/05/2026 - 09/05/2026

Organiser, Università della Svizzera Italiana, Lugano, Switzerland

Edutainment and Social 2025

31/05/2025 - 02/06/2025

Conference organiser, <https://ludicomix.it/edutainment-social/>
Empoli, Italy

Workshop: Theoretical Equivalence and Related Topics

08/04/2025

Chair, Università Svizzera Italiana,
Lugano, Switzerland

VII Urbino Summer School in Epistemology

25/08/2025 - 28/08/2025

Organiser, Università degli Studi di Urbino Carlo BO,
Urbino, Italy

XXVIII International Summer School in Philosophy of Physics - Epistemology of Quantum Gravity

26/05/2025 - 30/05/2025

Organiser, Università degli Studi di Urbino Carlo BO,
Urbino, Italy

MOOC - Tempo, Universo e Particelle. Una introduzione alla filosofia della fisica

10/2023 - 12/2024

Coordinator, Università degli Studi di Urbino Carlo BO,
Urbino, Italy

VI Urbino Summer School in Epistemology

27/08/2024 - 30/08/2024

Organiser, Università degli Studi di Urbino Carlo BO,
Urbino, Italy

XXVII International Summer School in Philosophy of Physics - Philosophy of Dark Energy

03/06/2024 - 07/06/2024

Member of the Organising Committee, Università degli Studi di Urbino Carlo BO,
Urbino, Italy

Edutainment and Social 2024

11/05/2024 - 12/05/2024

Conference organiser, <https://ludicomix.it/edutainment-social/>
Empoli, Italy

Reading Group Synergia

27/11/2023 - 31/03/2024

Reading Group organiser, Università degli Studi di Urbino Carlo BO,
Urbino, Italy

Edutainment and Social 2023

28/05/2023

Conference organiser, <https://www.ludicomix.it/edutainment-social/>
Empoli, Italy

MEMBERSHIP

from 20/01/2025 Member of the **European Philosophy of Science Association (EPSA)**

from 05/09/2024 Member of the **The Italian Society for Logic and the Philosophy of Science (SILFS)**

from 01/11/2023 Member of the **Synergia Research Group**

Editorial Board

Editor Scientific Committee: **Sophia: Research on Fundamentals and Correlation of Knowledge**

REFEERING

Journals: **Foundations of Science, Synthese, Isonomia, The Reasoner, Humana.mente**